APEX INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Experiment : 8

SUBJECT: Competitive Coding-1

SUBJECT CODE: 24CSP-291

SUBMITTED TO: Mr. Sourav Jaiswal

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SECTION: 24BDS-1(B)

Aim:

1. Best Time to Buy and Sell Stock

▸ Given array of stock prices (one price per day)

▸ Can buy once and sell once

▸ Must sell after buying (or not buy at all)

▸ Find maximum profit

Objectives:

- To understand how to **maximize profit using array traversal**
- To apply **greedy algorithm techniques**
- To track **minimum and maximum values dynamically**
- To solve problems using **single-pass optimization (O(n))**
- To improve skills in **decision making with constraints**

Pseudocode:

1. Best Time to Buy and Sell Stock

```
FUNCTION maxProfit(prices):

    minPrice = infinity
    maxProfit = 0

    FOR each price in prices:

        IF price < minPrice:
            minPrice = price

        ELSE:
            profit = price - minPrice
```

```
            maxProfit = MAX(maxProfit, profit)

    RETURN maxProfit
```

Code:

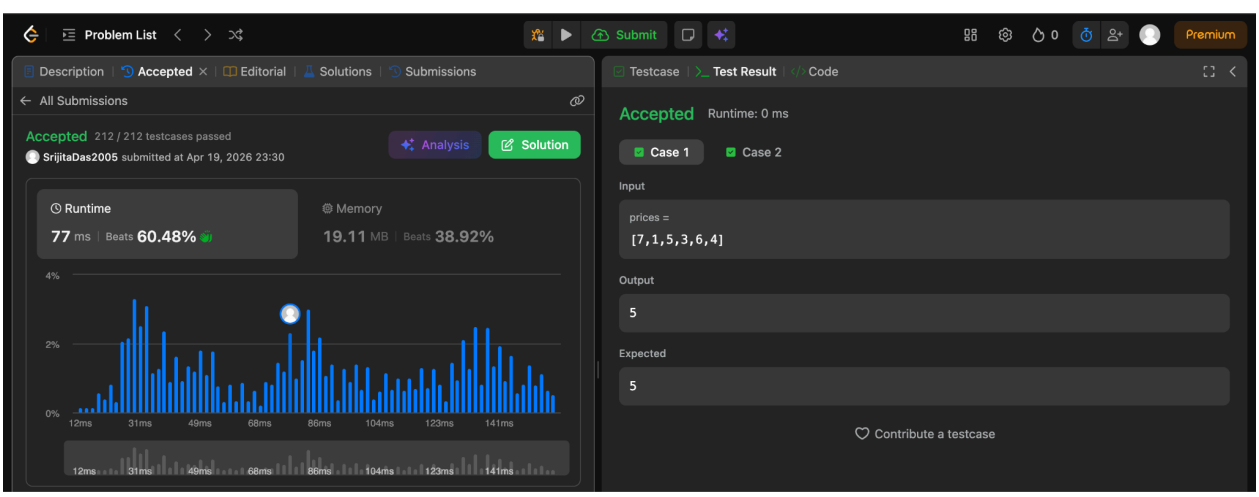
1.Best Time to Buy and Sell Stock

```
class Solution(object):
    def maxProfit(self, prices):
        min_price = float('inf')
        max_profit = 0

        for price in prices:
            if price < min_price:
                min_price = price
            else:
                max_profit = max(max_profit, price - min_price)

        return max_profit
```

Output:



Learning Outcome:

- **Identify optimal buy and sell points in stock data**
- **Use greedy approach to maximize profit**
- **Track minimum value efficiently while iterating**
- **Solve problems in O(n) time and O(1) space**
- **Avoid brute-force solutions and write optimized code**
- **Apply similar logic to real-world financial problems**